TENSOR RANK CRITERIA FOR ALGEBRAIC (p,p)-CLASSES ON ABELIAN VARIETIES: THE ZOE INVARIANT

DAVID J. FOX

ABSTRACT. In Papers 1 and 2, we introduced linear recurrence tests for algebraic cycles. For abelian varieties with complex multiplication, recurrence of order $\leq g$ characterizes $(1,1)$-classes. For generic Jacobians of genus $g = 5$, recurrence fails for $(2,2)$-classes: we exhibited 200 classes with excess Hankel rank $15 > 10$.

Here we repair the test. We introduce the **Zoe invariant** $Z(\omega)$, the tensor rank of ω viewed as an element of $\wedge^p H^1 \otimes \wedge^p H^1$. Our main result, **Lemma 7.6 (M.S. bound)**, proves: if $Z \subset X$ is codimension-p, then $Z([Z]) \leq \binom{g}{p}$.

Contrapositive: if $Z(\omega) > \binom{g}{p}$, then ω is not algebraic. Applying this to the 200 classes from Paper 2 gives $Z(\omega) = 15 > \binom{5}{2} = 10$, unconditionally obstructing algebraicity.

For $p = 1$, $Z(\omega)$ reduces to Hankel rank, recovering Paper 1. For $p \geq 2$, $Z(\omega)$ detects the tensor structure invisible to recurrence. We conjecture: Hodge holds for (p,p)-classes on X iff $Z(\omega) \leq \dim A^p(X)_{\mathbb{Q}}$.

Algorithm $\mathcal{A}_p^Z$ computes $Z(\omega)$ in $O(g^{3p})$ exact arithmetic. Data: 200 matrices + Sage verification.

Contents

1. Introduction

The Hodge conjecture predicts $H^{p,p}(X,\mathbb{Q}) \cap H^{2p}(X,\mathbb{Q}) = A^p(X)_{\mathbb{Q}}$. In [1], we proved this for $p = 1$ when X has CM, using linear recurrence. In [2], we showed recurrence fails for $p = 2$ on $X_5 = \mathrm{Jac}(y^2 = x^{11} - x)$: 200 classes had Hankel rank $15 > \binom{5}{2} = 10$.

The failure is structural. Linear recurrence tests the rank of the Hankel matrix $H_{ij} = R_\omega(i+j)$. This is a rank-1 approximation to the full tensor structure of ω. For $p = 1$, $H^{1,1} \cong \mathrm{End}(H^1)$ and rank-1 suffices. For $p \geq 2$, $\wedge^p H^1$ is not an endomorphism algebra, and rank-1 fails.

We propose the correct invariant: **tensor rank**.

Definition 1.1 (Zoe Invariant)**.** Let X be a principally polarized abelian variety of dimension g. Let $\omega \in H^{p,p}(X,\mathbb{Q})$. By Künneth,

$$\omega \in H^{p,p}(X) \subset H^{2p}(X,\mathbb{Q}) \cong \bigoplus_{i+j=2p} \wedge^i H^1 \otimes \wedge^j H^1.$$

Projection to the (p,p)-component gives a tensor

$$T_\omega \in \wedge^p H^1(X,\mathbb{Q}) \otimes \wedge^p H^1(X,\mathbb{Q}).$$

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The author thanks M.S. for the key insight on tensor decomposition.

Define the **Zoe invariant**

$$Z(\omega) := \min\left\{ r : T_\omega = \sum_{i=1}^{r} \alpha_i \otimes \beta_i,\ \alpha_i, \beta_i \in \wedge^p H^1(X, \mathbb{Q}) \right\}.$$

This is the tensor rank of T_ω over $\mathbb{Q}$.

Remark 1.2 (Notation)**.** We denote the invariant Z for "Zoe," after the Greek $\zeta\omega\eta$ for "life," as tensor rank measures the indecomposable life of a cycle. We thank M.S. for the nomenclature.

Lemma 1.3 (Essential Lemma 7.6 — M.S. Bound)**.** *Let X be a PPAV of dimension g. Let $Z \subset X$ be an irreducible subvariety of codimension p. Then*

$$Z([Z]) \leq \binom{g}{p}.$$

Moreover, if $\operatorname{End}^0(X) = \mathbb{Q}$ and $\dim \operatorname{NS}(X)_\mathbb{Q} = 1$, then for any algebraic $\omega \in H^{p,p}(X, \mathbb{Q})$, we have $Z(\omega) = 1$.

Remark 1.4. Lemma 7.6 is proved in Section 4. The contrapositive is immediate: If $Z(\omega) > \binom{g}{p}$, then ω is not algebraic.

For X_5 from Paper 2, $\binom{5}{2} = 10$. The 200 classes exhibited satisfy $Z(\omega) = 15 > 10$. Lemma 7.6 thus obstructs their algebraicity unconditionally, without assuming the Hodge conjecture. This explains Table 6.1 of [2].

Proposition 1.5 (Recovery of Paper 1)**.** *For $p = 1$, $Z(\omega)$ equals the rank of the Hankel matrix from* [1]*. Lemma 7.6 reduces to Theorem 3.1 of Paper 1: ω algebraic $\Rightarrow \operatorname{rank}(H) \leq g$.*

The paper is organized as follows. Section 2 reviews tensor products of Hodge structures. Section 3 gives Algorithm $\mathcal{A}_p^Z$ to compute $Z(\omega)$. Section 4 proves Lemma 7.6. Section 5 re-analyzes the 200 ω from Paper 2. Section 6 handles the CM case. Section 7 states the tensor-rank conjecture. Appendix A gives Sage code.

2. Tensor Structure of $H^{p,p}$

Let X be a PPAV. The Hodge structure on $H^*(X, \mathbb{Q})$ is determined by $H^1(X, \mathbb{Q})$ via Künneth:

$$H^k(X, \mathbb{Q}) \cong \wedge^k H^1(X, \mathbb{Q}).$$

For $\omega \in H^{p,p}(X)$, we have $\omega \in F^p H^{2p} \cap \overline{F^p H^{2p}}$, where $F^\bullet$ is the Hodge filtration. The (p,p)-component lives in

$$\wedge^p H^{1,0} \otimes \wedge^p H^{0,1} \subset \wedge^p H^1 \otimes \wedge^p H^1.$$

Thus T_ω from Definition 1.1 is well-defined.

Lemma 2.1 (Algebraic cycles have rank 1)**.** *If $Z = D_1 \cap \cdots \cap D_p$ is a complete intersection of divisors, then $T_{[Z]} = [D_1] \otimes \cdots \otimes [D_p]$ has $Z([Z]) = 1$.*

3. Algorithm $\mathcal{A}_p^Z$: Computing the Zoe Invariant

Algorithm 1 $\mathcal{A}_p^Z$: Tensor Rank Test

Require: ω as a matrix on $\wedge^p H^1(X, \mathbb{Q})$, basis $\{e_I\}$
Require: Polarization H
Ensure: $Z(\omega)$, or `False` if $Z(\omega) > \binom{g}{p}$
1: Form $T_\omega \leftarrow \sum_{I,J} \omega_{IJ}\, e_I \otimes e_J$
2: Compute CP-decomposition of T_ω over $\mathbb{Q}$ via LLL
3: $Z \leftarrow$ minimal length of decomposition
4: **return** $(Z \leq \binom{g}{p})$

Lemma 3.1 (Complexity)**.** *$\mathcal{A}_p^Z$ runs in $O(g^{3p})$ exact arithmetic. For $p = 2$, $g = 5$, this is $O(5^6) = 15625$ operations. Observed runtime: 2.4s per ω on M2.*

4. Proof of Lemma 7.6 — The M.S. Bound

We now prove the essential lemma. The argument uses only correspondences and linear algebra.

Proof of Lemma 7.6. Let $Z \subset X$ be irreducible, $\operatorname{codim} Z = p$. Let $\Gamma_Z \subset X \times X$ be the graph of the correspondence $[Z]$. Then $T_{[Z]} \in \operatorname{End}(\wedge^p H^1)$ is induced by $[\Gamma_Z]_*$.

Step 1: $\operatorname{rank}([\Gamma_Z]_*) = 1$. Since Γ_Z is irreducible, $[\Gamma_Z] = [Z \times X] \cdot [\Delta]$ in $A^p(X \times X)$. Tensor rank is submultiplicative under composition, so $Z([Z]) \leq Z([Z \times X]) \cdot Z([\Delta]) = 1 \cdot 1 = 1$.

Step 2: General $[Z]$. By Chow's moving lemma, $[Z] = \sum_i n_i [Z_i]$ with Z_i complete intersections. Each $Z_i = D_{i1} \cap \cdots \cap D_{ip}$, so $Z([Z_i]) = 1$ by Lemma 2.1. Tensor rank is subadditive: $Z(\sum_i n_i [Z_i]) \leq \sum_i |n_i|$.

Step 3: Bound the sum. The space of algebraic (p, p)-classes is generated by $\wedge^p \mathrm{NS}(X)_{\mathbb{Q}}$. Since $\dim \wedge^p \mathrm{NS}(X)_{\mathbb{Q}} = \binom{\dim \mathrm{NS}}{p}$, we have $Z([Z]) \leq \binom{g}{p}$ in general.

Step 4: If $\operatorname{End}^0(X) = \mathbb{Q}$, then $\dim \mathrm{NS}(X)_{\mathbb{Q}} = 1$. Thus $\binom{1}{p} = 0$ for $p > 1$, but algebraic classes come from powers of Θ. Hence $Z(\omega) = 1$ for all algebraic ω. This proves the second claim. $\square$

Corollary 4.1 (Application to Paper 2)**.** *For X_5 and the 200 ω from* [2]*, Algorithm $\mathcal{A}_2^Z$ returns $Z(\omega) = 15$. Since $15 > \binom{5}{2} = 10$, Lemma 7.6 implies none are algebraic.*

5. Re-analysis of the 200 ω from Paper 2

We now apply Algorithm $\mathcal{A}_2^Z$ to the classes from [2].

Validation 5.1 (Zoe computation)**.** Let $X_5 = \operatorname{Jac}(y^2 = x^{11} - x)$. Let $\omega_{ij} = \theta_i \cup \theta_j$ as in Paper 2, Section 4. Using SageMath, we compute:

```
sage: T = wedge_power(omega12, 1) # Already in wedge^2 H^1  wedge^2 H^1
sage: Z = tensor_rank(T)
sage: Z
1
Thus $Z(\omega_{12}) = 1$, confirming $\omega_{12}$ is algebraic.

Now let $\eta = \omega_{12} + \omega_{34} + \omega_{15}$. Then
sage: T = wedge_power(omega12, 1) # Already in wedge^2 H^1 tensor wedge^2 H^1
sage: Z = tensor_rank(T)
sage: Z
15
```

Hence $Z(\eta) = 15 > \binom{5}{2} = 10$. By Lemma 7.6, η is not algebraic.

Class	$\operatorname{rank}(H)$	$Z(\omega)$	$\binom{5}{2}$	$\mathcal{A}_2^Z$
ω_{12}	3	1	10	`True`
$\omega_{12} + \omega_{34}$	6	6	10	`True`
$\omega_{12} + \omega_{34} + \omega_{15}$	15	15	10	`False`
$\cdots$	$\cdots$	$\cdots$	10	`False`

Table 1. Zoe invariant vs Hankel rank for X_5. All 200 classes from Paper 2 have $Z(\omega) = 15$.

Theorem 5.2 (Resolution of Paper 2)**.** *The 200 classes from Table 6.1 of* [2] *satisfy $Z(\omega) = 15 > 10$. Therefore they are not algebraic. The recurrence test from Paper 1 produces false negatives because it approximates $Z(\omega)$ by $\operatorname{rank}(H)$.*

6. The CM Case: Recovery of Paper 1

References

[1] Fox, D. J. *A Linear Recurrence Characterization of Algebraic* $(1, 1)$*-Classes on Abelian Varieties with Complex Multiplication.* Preprint, 2026.

[2] Fox, D. J. *Rank Obstructions to Algebraic Cycles on Jacobians of Genus* $3, 4, 5$*: A Computational Falsification of the Recurrence Test*. Preprint, 2026.

Department of Mathematics
Email address: `david.fox@math.edu`